

The Ferrand Family Office

The pitch was for an asset that would lose to the existing portfolio on return, lose on volatility, and lose on almost every measure the investment committee normally used, and Théo Ferrand wanted to buy it.

The Ferrand family office managed \$840 million across public equities, private credit, and a small direct real-estate book. The blended portfolio had a long-run expected return of 7.8 percent and a standard deviation of 12.4 percent. The risk-free rate was 3.5 percent, so the portfolio's Sharpe ratio was 0.347.

The candidate was a managed futures fund — a systematic trend-following program run by Ashwood Systematic, with an expected return of 5.2 percent and a standard deviation of 15.0 percent. On its own it was strictly worse than the existing portfolio: less return, more risk, a Sharpe ratio of 0.113. Ashwood charged 1.5 percent on assets and 15 percent of profits, which was most of a percentage point of the expected return in a typical year.

What the fund had was a correlation with the Ferrand portfolio of negative 0.15, estimated over eighteen years including three drawdowns.

Two members of the committee had objected in writing. The first objection was that a 5.2 percent expected return did not clear the family's 6 percent spending policy plus inflation, so the position could not pay its way. The second was that the correlation estimate was a historical artifact — trend followers had been negatively correlated to equities in the crises everyone remembered precisely because those crises were slow, and in a fast one the correlation would go to zero or worse.

Ferrand's own view was that the committee was arguing about the wrong thing. "Nobody is proposing we hold it by itself," he said. "The question is whether the portfolio that contains it is better than the portfolio that does not, and that is arithmetic, not opinion. What is opinion is the correlation. So let us do the arithmetic first and then argue about the one input that deserves it."

EXHIBIT 1 — INPUTS

	EXISTING PORTFOLIO	ASHWOOD SYSTEMATIC
Expected return	7.8%	5.2%
Standard deviation	12.4%	15.0%
Correlation with existing portfolio	—	(0.15)

Risk-free rate: 3.5 percent. Ashwood's expected return is stated net of fees. Short positions in either asset are not permitted.

ASSIGNMENT

- 1 There is a one-line rule for whether adding a new asset to a portfolio improves the Sharpe ratio. Derive it or state it, and apply it to Ashwood.
- 2 Compute the tangency portfolio of the two assets. What weight does the optimizer put on the asset that is worse on every standalone measure?
- 3 Report the expected return, standard deviation, and Sharpe ratio of the blended portfolio. Compare to the existing portfolio.
- 4 Answer the first objection: the position does not clear the spending policy. Is that a reason not to hold it?
- 5 Answer the second objection by finding the number that settles it. At what correlation does Ashwood stop improving the portfolio? Separately, at what net expected return does it stop improving the portfolio at the estimated correlation of negative 0.15?